



Ethics & Bias

CSE472: HCI – Human Computer Interaction

Lecture – 18: Bias and Ethics in
HCI

Spring 2026



Inspiring Excellence

What Are Ethics?

- The study of moral standards and how they affect conduct
- Moral standards are ...
 - A system of principles governing the appropriate conduct of an individual
- Ethics ask us to live “mindfully”

ACM Code of Ethics

General Moral Imperatives

1. Contribute to society and human well-being
2. Avoid harm to others
3. Be honest and trustworthy
4. Be fair and take action not to discriminate
5. Honor property rights including copywrites and patents
6. Give proper credit for intellectual property
7. Respect the privacy of others
8. Honor confidentiality

More Specific Professional Responsibilities

Strive to achieve the highest quality, effectiveness, and dignity in both the process and products of professional work

1. Acquire and maintain professional competence
2. Know and respect existing laws pertaining to professional work
3. Accept and provide appropriate professional review
4. Give comprehensive and thorough evaluations for computer systems and their impacts, including analysis of possible risks
5. Honor contracts agreements, and assigned responsibilities
6. Improve public understanding of computing and its consequences
7. Access computing and communication resources only when authorized to do so

Why Should You Care about Ethics?

- As designers, engineers, researchers you have great power
 - People will believe that you are an expert
 - People will do what you tell them
- People will blame themselves
 - For errors, mistakes
 - When things go wrong
 - For not giving you the results you want
 - Participants might be reduced to tears, or worse...
- With great power comes great responsibility



Milgram Experiment

- Psychology experiment at Yale
- Prof. Stanley Milgram
- Study of obedience, willingness to obey authority (1963)
 - Connection to WW2



Public Announcement

**WE WILL PAY YOU \$4.00 FOR
ONE HOUR OF YOUR TIME**

Persons Needed for a Study of Memory

*We will pay five hundred New Haven men to help us complete a scientific study of memory and learning. The study is being done at Yale University.

*Each person who participates will be paid \$4.00 (plus 50c carfare) for approximately 1 hour's time. We need you for only one hour: there are no further obligations. You may choose the time you would like to come (evenings, weekdays, or weekends).

*No special training, education, or experience is needed. We want:

Factory workers	Businessmen	Construction workers
City employees	Clerks	Salespeople
Laborers	Professional people	White-collar workers
Barbers	Telephone workers	Others

All persons must be between the ages of 20 and 50. High school and college students cannot be used.

*If you meet these qualifications, fill out the coupon below and mail it now to Professor Stanley Milgram, Department of Psychology, Yale University, New Haven. You will be notified later of the specific time and place of the study. We reserve the right to decline any application.

*You will be paid \$4.00 (plus 50c carfare) as soon as you arrive at the laboratory.

TO:

PROF. STANLEY MILGRAM, DEPARTMENT OF PSYCHOLOGY,
YALE UNIVERSITY, NEW HAVEN, CONN. I want to take part in
this study of memory and learning. I am between the ages of 20 and
50. I will be paid \$4.00 (plus 50c carfare) if I participate.

NAME (Please Print)

ADDRESS

TELEPHONE NO. Best time to call you

AGE OCCUPATION SEX

CAN YOU COME:

WEEKDAYS EVENINGS WEEKENDS

Milgram Experiment

- 3 roles:
 - Teacher (Participant)
 - Learner
 - Experimenter (Professor)
- Teacher & learner in separated rooms
 - Able to communicate, but not see each other
- Teacher read word pairs, learner had to remember them
 - If answer incorrect teacher had to administer electric shock to learner
 - Teacher received real electric shock as proof



Milgram Experiment

- Teacher believed that learner received shocks (15V-400V)
- BUT in reality:
 - No shocks
 - Not the learner who responded
 - Tape recorder integrated in the electro-shock generator
- Experiment only stopped if:
 - Teacher (really) wished to stop
 - After giving 450 V three times (potentially lethal)

Milgram Experiment

- 26 out of 40 (65%) of participants administered final 450 V shock (potentially lethal!)
- Only 5 stopped before 300 volts
- Even if teachers wanted to stop, only a few had resources to resist authority (experimenter)
- None of the teachers who wanted to stop:
 - Insisted that experiment should be terminated
 - Checked learner's health

Milgram Experiment

- Very stressful for participants
- Often protested, but still obeyed
- Milgram: "There were powerful reactions of tension and emotional strain in a substantial proportion of the participants. Persons were observed to sweat, tremble, stutter, bite their lips..."
- Ordinary people can become agents in a terrible destructive process



Milgram Experiment

- Many have questioned the ethics of the study
 - Milgram defended his work
 - Participants were debriefed
 - Many reported positive experiences
 - Claimed benefits outweighed any harm

What do you think?

Are Ethics Still Relevant Today?



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What Ethical Issues Might You Face?

- People will do what you tell them to do!
- The boss asks you to create an app that will violate people's privacy
- You need to stop supporting a service that people have become dependent on
- You to cut corners in design/testing to save money
- You don't have permission to collect people's data but you do it anyway (or trick them into giving you permission)

How Can You Do Ethical Design?

- Thorough design
- Robust implementation
- Honesty in advertising
- Thorough risk analysis
- Thorough testing
- Proper training
- Take it seriously. There are numerous examples of bad design that have led to harm and even death!

Summary of Ethics

- As designers, engineers, researchers you have great power
 - People will do what you tell them
- People will blame themselves
 - For errors, mistakes
 - When things go wrong
 - For not giving you the results you want
 - Participants might be reduced to tears, or worse...
- With great power comes great responsibility
 - IRB, ethics review, etc. helps but is only part of the story
 - It's up to you to be responsible for your own conduct

Bias

What is bias?

Prejudice in favor of or against one thing, person, or group compared with another, usually in a way considered to be unfair.

- Expressed directly: "I like rich people more than poor people."
- Expressed indirectly: E.g., Sitting further away from a poor person than a rich person.

Types of Bias

- Explicit
 - Person is aware of his/her evaluation
 - Conscious bias
- Implicit
 - Person doesn't perceive or endorse evaluation
 - Subconscious bias
- Sure, other people are biased, but not me...
 - Lots of evidence that shows ALL people are biased, at least subconsciously
 - Even against their own group. i.e. women are biased against women, black people are biased against black people, etc.

Types of Bias You Should Think About

- Selection bias
- Social desirability bias
- Demand characteristics
- Evaluator/Interviewer bias
- Acquiescing
- Extreme responding

...

Other subconscious biases

Selection Bias

- Selection of individuals, groups or data in a way that proper randomization is not achieved.
- The sample obtained is not representative of the population intended to be analyzed.

e.g., most psychology experiments done with “WEIRD” people

- Western, educated, industrialized, rich and democratic
 - (i.e. US college students)
- Not representative of the entire world / population

Selection Bias ~ Examples

- Online surveys on a website
 - e.g. a website devoted to preventing harassment of women concluded that nearly all women were victims of harassment.
- Online Reviews of restaurants, films, etc.
 - May be biased towards negative because people are more likely to rant after having a bad time than they are to rave after a good or adequate time.
- Self selection
 - People who choose to participate are likely to be people who are already invested

Social Desirability Bias

Broadly conceived, 'social desirability' as a response determinant refers to the tendency of people to deny socially undesirable traits or qualities and to admit to socially desirable ones.

Phillips and Clancy (1972)

Lots of research that shows:

- people will say they voted when they haven't
- will inflate their reported incomes
- will describe themselves in socially-sanctioned ways (such as saying they are happier than they are in their marriages)
- will edit politically "incorrect" responses (such as prejudice)
- etc.

Demand Characteristics

- Subtle cues that make participants aware of what the experimenter expects to find or how participants are expected to behave.
- Participants alter their behavior to conform to expectations.
 - People want to be “good participants”.
 - People want to “help” the researcher.
 - People want to contribute to research, science, society, etc.
- Even if you try to hide the purpose of a study, people will guess the purpose and conform to what they guessed.

Interviewer Bias

- Is a type of research bias that occurs when the interviewer unintentionally influences a participant's responses, leading to distorted or inaccurate data.
- Results/data varies depending on the interviewer.
- Variations in both how participants AND interviewers behave.

Acquiescing and Extreme Responding

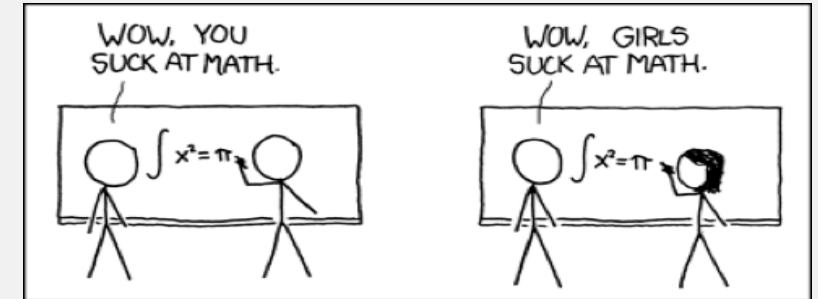
- **Acquiescing**: tendency to agree with all the questions.
 - People like to be agreeable.
 - Especially prevalent if survey/interview questions are “leading”.
 - Did you have an enjoyable experience?
 - Would you recommend my product to others?
- **Extreme Responding**: tendency to select most extreme options.
 - General indifference
 - Willingness to please
 - Poor question wording

Strongly Disagree	Disagree	Undecided	Agree	Strongly Agree
(1)	(2)	(3)	(4)	(5)

Subconscious Biases

- Many... (perceived) wealth, education, tallness, etc.

- Race
- Gender



- Subconscious bias affects everything, e.g.,
 - Jobs, employment, recruiting, salary
 - Course evaluations, performance evaluations
 - Negotiations - mortgages, car sales, etc.
 - Law enforcement
 - And so on...

Subconscious Biases

'Social sciences don't have laws, but if we were attempting to devise one, "Women and minorities get lower teaching evaluations" would be pretty close to axiomatic.'

Bias Against Female Instructors

| New analysis offers more evidence against the reliability of student evaluations of teaching.

INSIDE
HIGHER ED

Teaching evaluations are often used to confirm the worst stereotypes about women faculty (opinion)

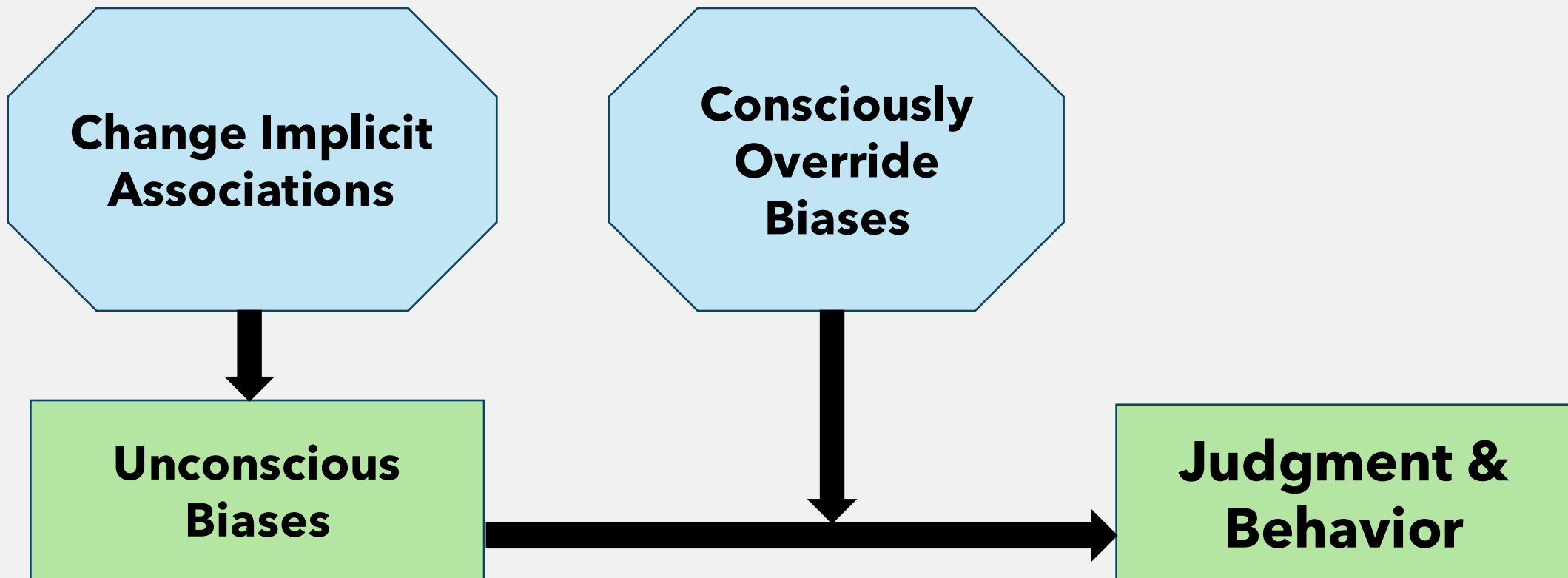
Such evaluations pretend to be the result of a neutral process but are better measures of student...

INSIDEHIGHERED.COM

Best Way for Professors to Get Good Student Evaluations? Be Male.

What Can You Do about Bias?

- So, you can't believe anything that participants say.
 - Should you just give up?
- A Two Strategy Solution.



1. Overriding Unconscious Bias

- Be aware of the potential for bias
- Be motivated to control bias
- Take the time to consider individual characteristics and avoid stereotyped evaluations

2. Change implicit assumptions

- Education
 - Seminars and classes on implicit bias
 - Media articles
 - Books
- Exposure
 - Hang out with people different from you
 - Attend and promote diversity events
 - Make sure your teams etc. are diverse
 - etc.

Take~Aways

- Implicit bias is distinct from conscious motivation
- We all have these biases due to cultural exposure
- They can affect behavior: we need to work to override them
- They can be changed over time with education and exposure
- Most important: Recognize and accept that you (and everyone) are biased! Pay attention to when bias may affect your life, work, choices, etc.

Any Questions?